Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

When aluminium is bombarded with fast neutrons, it changes into sodium with emission of particle x according to the equation
$^{27}_{13}\mathrm{Al} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{24}_{11}\mathrm{Na} + x$
What is x?

(A) electron

(B) proton

(C) neutron

(D) alpha-particle

Q.2

The binding energies per nucleon for $^{2}_{1}\mathrm{H}$ and $^{4}_{2}\mathrm{He}$ respectively are $1.1\mathrm{MeV}$ and $7.1\mathrm{MeV}$. The energy released (in MeV) when two $^{2}_{1}\mathrm{H}$ nuclei fuse to form $^{4}_{2}\mathrm{He}$ is

(A) 4.4

(B) 8.2

(C) 24

(D) 28.4

Q.3

A carbon nucleus emits a particle x and changes into nitrogen according to the equation
$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + x$
What is x?

(A) an electron

(B) a proton

(C) an alpha particle

(D) a photon

Q.4

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The binding energy per nucleon of $\mathrm{C}-12$ is $7.68\ \mathrm{MeV}$ and of $\mathrm{C}-13$ is $7.48\mathrm{MeV}$. The energy (in MeV) required to remove the extra neutron from C-13 is very nearly equal to

(A) 0.2

(B) 3.7

(C) 3.9

(D) 5

Q.5

A freshly prepared radioactive source of half life $2\ \mathrm{h}$ emits radiation of intensity which is 64 times the permissible safe level. The minimum time after which it would be possible to work safely with this source is

(A) $6\ \mathrm{h}$

(B) $12\ \mathrm{h}$

(C) $24\ \mathrm{h}$

(D) $128\ \mathrm{h}$

Q.6

The activity of a radioactive element reduces to $\frac{1}{16}$ of its original value in 36 years. The half life of the element is

(A) 6 years

(B) 9 years

(C) 12 years

(D) 18 years

Q.7

The transition from state $n=4$ to $n=3$ in a hydrogen like atom results in ultraviolet radiation. Infrared radiation will be obtained in the transition

(A) $2 \rightarrow 1$

(B) $3 \rightarrow 2$

(C) $4 \rightarrow 2$

(D) $5 \rightarrow 4$

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Q.8

A radioactive element of mass number 208 at rest disintegrates by emitting an α-particle. If E is the energy of the emitted α-particle, the energy of disintegration is

(A) $\frac{52}{51}E$

(B) $\frac{51}{52}E$

(C) $52E$

(D) E

Q.9

If the radius of a nucleus $^{256}\mathrm{X}$ is 8 fermi, the radius of ^{4} He nucleus will be

(A) 1 fermi

(B) 2 fermi

(C) 3 fermi

(D) 4 fermi

Q.10

The binding energy of deuteron $\left({}_1^2\mathrm{H}\right)$ is $1.15\mathrm{MeV}$ per nucleon and an alpha particle $\left({}_2^4\mathrm{He}\right)$ has a binding energy of $7.1\mathrm{MeV}$ per nucleon. Then in the reaction ${}_1^2\mathrm{H}+{}_1^2\mathrm{H} \rightarrow {}_2^4\mathrm{He}+\mathrm{Q}$ the energy Q released is

(A) $1\mathrm{MeV}$

(B) $11.9\mathrm{MeV}$

(C) $23.8\mathrm{MeV}$

(D) $931\mathrm{MeV}$

Q.11

A star initially has 10^{40} deuterons. It produces energy via the processes
${}_1^2\mathrm{H}+{}_1^2\mathrm{H} \longrightarrow {}_1^3\mathrm{H}+p$
${}_1^2\mathrm{H}+{}_1^3\mathrm{H} \longrightarrow {}_2^4\mathrm{He}+n$
where the masses of the nuclei are : $m\left({}^2\mathrm{H}\right)=2.014\mathrm{u}\, m(p)=1.007\mathrm{u}, m(n)=1.008\mathrm{u}$ and $m\left({}^4\mathrm{He}\right)=4.001\mathrm{u}$ If the average power radiated by the star is 10^{16} W, the deuteron supply of the star is exhausted in a time of the order of

(A) 10^6 s

(B) 10^8 s

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(C) 10^{12} s

(D) 10^{16} s

Q.12

During a negative beta decay:

(A) an atomic electron is ejected

(B) an electron which is already present within the nucleus is ejected

(C) a neutron in the nucleus decays emitting an electron

(D) a part of the binding energy of the nucleus is converted into an electron.

Q.13

The radioactivity of a sample is X at a time t_1 and Y at a time t_2. If the mean life of the specimen is τ, the number of atoms that have disintegrated in the time interval $(t_2 - t_1)$ is

(A) $Xt_1 - Yt_2$

(B) $X - Y$

(C) $(X - Y)/\tau$

(D) $(X - Y)\tau$

Q.14

A nucleus at rest splits into two nuclear parts having radii in the ratio $1 : 2$. Their velocities are in the ratio

(A) $8 : 1$

(B) $6 : 1$

(C) $4 : 1$

(D) $2 : 1$

Q.15

The energy released by the fission of one uranium atom is 200MeV. The number of fissions per second required to produce $3.2\ \text{W}$ of power is

(A) 10^7

(B) 10^{10}

(C) 10^{15}

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(D) 10^{11}

Q.16

The activity of a freshly prepared radioactive sample is 4×10^{10} disintegrations per second. The mean life of the sample is 10^{9} s and the mass of an atom of this radioisotope is 10^{-25} kg. The mass of the radioactive sample is

(A) 1mg

(B) 2mg

(C) 4mg

(D) 8mg

Q.17

The half-life of $^{131}\mathrm{I}$ is 8 days. Given a sample of $^{131}\mathrm{I}$ at time $t = 0$, we can assert that

(A) no nucleus will decay before $t = 4$ days

(B) no nucleus will decay before $t = 8$ days

(C) all nuclei will decay before $t = 16$ days

(D) a given nucleus may decay any time after $t = 0$.

Q.18

Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $1/$ e after a time

(A) $\frac{1}{10\lambda}$

(B) $\frac{1}{11\lambda}$

(C) $\frac{11}{10\lambda}$

(D) $\frac{1}{9\lambda}$

Q.19

After two hours $1/16$ th of the initial amount of a certain radioactive isotope remains undecayed. The half life of the isotope is

(A) 15 min

(B) 30 min

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(C) 45 min

(D) 1 hour

Q.20

The electron emitted in beta radiation originates from

(A) inner orbits of atoms

(B) free electrons existing in nuclei

(C) decay of a neutron in a nucleus

(D) photon escaping from the nucleus

Q.21

The activity of a radioactive sample A is the half that of a sample B. If sample A has twice the number of nuclei than sample B, the ratio of the half life of A to that of B is

Q.22

The electron in a hydrogen atom makes a transition from a higher state n_1 to a lower state n_2. The radius of the electron orbit in state n_1 is 4 times that in state n_2.. Find n_1 and n_2. Find the ratio of $\frac{n^1}{n^2}$

Q.23

The half-life of a substance is 20 minutes. What is the time interval between 33% decay and 67% decay? Find time interval T divided by 10

Q.24

The radius of the electron orbit in the ground state of hydrogen is $0.53\tilde{A}\ldots$. Due to collision with a neutron, the electron shifts to a state in which its orbital radius becomes $4.77\tilde{A}\ldots$. The principal quantum number of the final state of hydrogen is

Q.25

The half life of a radioactive sample is 6.93 days. After how many months will only one-twentieth of the sample be left over? Take $\log_e(20) = 3.0$

Answers & Solutions

Q.1 Answer:

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alpha-particle

Solution:

The mass number of $x = 27 + 1 - 24 = 4$ and its atomic number $= 13 + 0 - 11 = 2$. Hence particle x is helium nucleus, which is called alpha particle.

Q.2 Answer:

24

Solution:

$_1^2\mathrm{H} + _1^2\mathrm{H} \rightarrow _2^4\mathrm{He} + Q$

Binding energy of $_1^2\mathrm{H} = 2 \times 1.1 = 2.2\mathrm{MeV}$. Therefore, the binding energy of two $_1^2\mathrm{H} = 2.2 + 2.2 = 4.4\ \mathrm{MeV}$. Now, the binding energy of $_2^4\mathrm{He}$ nucleus = $4 \times 7.1 = 28.4\mathrm{MeV}$. Hence $Q = 28.4 - 4.4 = 24\mathrm{MeV}$ which is choice (c).

Q.3 Answer:

an electron

Solution:

The charge of x is opposite to that of a proton. Hence x is an electron.

Q.4 Answer:

5

Solution:

Energy required $= (7.48 \times 13 - 12 \times 7.68)\mathrm{MeV} = 5.08\mathrm{MeV}$. Hence the correct choice is (d).

Q.5 Answer:

$12\ \mathrm{h}$

Solution:

The radioactivity of a sample decreases to $\left(\frac{1}{2^n}\right)$ in n half lives. Since the initial activity is 64 times the permissible level, it must become $1/64$ th of its initial value.

$\therefore \quad \frac{1}{64} = \frac{1}{2^n} \quad \text{or} \quad n = 6$

In other words in six half-lives the activity will reduce itself to the permissible level. Since half-life is 2 hours, the total time taken $= 2 \times 6\ \mathrm{h} = 12$ hours.

Q.6 Answer:

9 years

Solution:

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Activity at a given time is proportional to the number of radioactive atoms present at that time. The number of atoms left undecayed after 36 years = $\frac{1}{16}$ of the initial value, i.e. $N = N_0/16$ From $\frac{N}{N_0} = \left(\frac{1}{2}\right)^n$, we get $n = 4$. Now $t = nT_{1/2}$ Hence

$T_{1/2} = \frac{t}{n} = \frac{36 \text{ years}}{4} = 9 \text{ years}$

Hence the correct choice is (b).

Q.7 Answer:

$5 \rightarrow 4$

Solution:

The frequency of infrared radiation is less than that of ultraviolet radiation. Hence, infrared radiation has higher energy than ultraviolet radiation. Therefore, infrared radiation will be obtained in the transition $n = 5$ to $n = 4$, which is choice (d).

Q.8 Answer:

$\frac{52}{51}E$

Solution:

Mass number of daughter nucleus $(M) = 208 - 4 = 204$

Now, total energy of disintegration = energy of daughter nucleus + energy of α-particle

or $E_t = \frac{p^2}{2m} + \frac{P^2}{2M}$

Since momentum is conserved, $p = P$. Hence

$E_t = \frac{p^2}{2}\left(\frac{1}{m} + \frac{1}{M}\right) = \frac{p^2}{2}\left(\frac{m+M}{mM}\right)$

Energy of α-particle $= \frac{p^2}{2m} = E$. Hence

$$E_t = \left(\frac{m+M}{M}\right)E$$

$$= \left(\frac{4+204}{204}\right)E = \frac{52}{51}E$$

Hence the correct choice is (a).

Q.9 Answer:

2 fermi

Solution:

$R = R_0(A)^{1/3}$. Therefore

$\frac{R_2}{R_1} = \left(\frac{A_2}{A_1}\right)^{1/3} = \left(\frac{4}{254}\right)^{1/3} = \frac{1}{4}$

$R_2 = \frac{R_1}{4} = \frac{8 \text{ fermi}}{4} = 2$ **fermi**

Hence the correct choice is (b).

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Q.10 Answer:

$23.8\mathrm{MeV}$

Solution:

Binding energy of ${}_1^2\mathrm{H} = 1.15\times$ **number of nucleons** $= 1.15 \times 2 = 2.3\mathrm{MeV}$**. Total binding energy of reactants** $= 2.3 + 2.3 = 4.6\mathrm{MeV}$**. Binding energy of** ${}_2^4\mathrm{He} = 7.1\times$ **number of nucleons** $= 7.1 \times 4 = 28.4\mathrm{MeV}$ **Therefore,** $Q = 28.4 - 4.6 = 23.8\mathrm{MeV}$**. Hence the correct choice is (c).**

Q.11 Answer:

10^{12} s

Solution:

Let the mass of ${}_1^3\mathrm{H}$ **be** x **amu. Then the mass defect in the first process is**

$$(\Delta m)_1 = m\left({}_1^2\mathrm{H}\right) + m\left({}_1^2\mathrm{H}\right) - m\left({}_1^3\mathrm{H}\right) - m(p)$$
$$= 2.014 + 2.014 - x - 1.007 = (3.021 - x)\mathrm{u}$$

Mass defect in the second process will be

$$(\Delta m)_2 = m\left({}_1^2\mathrm{H}\right) + m\left({}_1^3\mathrm{H}\right) - m\left({}_2^4\mathrm{H}\right) - m(n)$$
$$= 2.014 + x - 4.001 - 1.008 = (x - 2.995)\mathrm{u}$$

$\therefore$ Total mass defect is

$$\Delta m = (\Delta m)_1 + (\Delta m)_2 = (3.021 - x) + (x - 2.995)$$
$$= 0.026\mathrm{u}$$

Now $1\mathrm{amu} = 931\mathrm{MeV} = 931 \times 10^6 \times 1.6 \times 10^{-19}$ J **Hence, energy released is given by**

$$E = 0.026 \times 931 \times 10^6 \times 1.6 \times 10^{-19}$$
$$= 3.87 \times 10^{-12}\ \mathrm{J}$$

Now power $P = 10^{16}$ W**. Therefore, the number of deutrons per second is** $\frac{10^{16}}{3.87\times10^{-12}} = 2.58 \times 10^{27}$ **per second**

$\therefore$ Deutron supply will exhaust in $\frac{10^{40}}{2.58\times10^{27}} = 3.87 \times 10^{12}$ **seconds**

Hence the correct choice is (c).

Q.12 Answer:

a neutron in the nucleus decays emitting an electron

Solution:

Negative β **decay is expressed by the equation:**

$n \longrightarrow p + e^- + \bar{v}$; $\bar{v}$ **stands for antineutrino.**

Hence the correct choice is (c).

Q.13 Answer:

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$(X-Y)\tau$

Solution:

Activity $\frac{dN}{dt} = -\lambda N$

$\therefore x = -\lambda N_{t_1}, y = -\lambda N_{t_2}$

$\Rightarrow N_{t_1} = -\frac{x}{\lambda}, N_{t_2} = -\frac{y}{\lambda}$

The number of atoms disintegrated in time interval

$\Rightarrow N_{t_2} - N_{t_1} = -\frac{y}{\lambda} - \left(-\frac{x}{\lambda}\right)$

$= \frac{x-y}{\lambda} = (x-y)\tau$

Q.14 Answer:

$8:1$

Solution:

Let A_1 **and** A_2 **be the mass numbers of the two nuclear parts. Their radii are given by**

$R_1 = R_0 (A_1)^{1/3}$ and $R_2 = R_0 (A_2)^{1/3}$

Dividing, we get

$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3}$ or

$\frac{A_1}{A_2} = \left(\frac{R_1}{R_2}\right)^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

Hence the ratio of their masses is

$\frac{m_1}{m_2} = \frac{1}{8}$

From the principle of conservation of momentum, the magnitude of p_1 **= magnitude of** p_2 **or** $m_1 v_1 = m_2 v_2$ **which gives**

$\frac{v_1}{v_2} = \frac{m_2}{m_1} = \frac{8}{1}$**, which is choice (a).**

Q.15 Answer:

10^{11}

Solution:

Energy released $= 200\text{MeV} = 200 \times 10^6 \text{eV} = 200 \times 10^6 \times 1.6 \times 10^{-19}$ **J. Therefore, the required number is** $\frac{3.2}{200 \times 10^6 \times 1.6 \times 10^{-19}} = 10^{11}$

Q.16 Answer:

4mg

Solution:

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Activity $|R| = \lambda N = \frac{N}{\tau}$ **where** $\tau = \frac{1}{\lambda}$ **is the mean life. Therefore,**

$$N = \tau|R|$$
$$= 10^9 \times \left(4 \times 10^{10}\right)$$
$$= 4 \times 10^{19}$$

Total mass of sample $= \left(4 \times 10^{19}\right) \times 10^{-25}$

$= 4 \times 10^{-6}$ kg $= 4$mg

So the correct choice is (c).

Q.17 Answer:

a given nucleus may decay any time after $t = 0$.

Solution:

A radioactive nucleus may decay any time after $t = 0$**. The number of nuclei at any time** t **is given by** $N = N_0 e^{-\lambda t}$**, where** $\lambda = \frac{0.693}{T}$ **where** T **is the half-life. Hence the correct choice is (d).**

Q.18 Answer:

$\frac{1}{9\lambda}$

Solution:

$N_1 = N_0 e^{-\lambda_1 t}, N_2 = e^{-\lambda_2 t}$**. Therefore**

$\frac{N_1}{N_2} = \frac{e^{-\lambda_1 t}}{e^{-\lambda_2 t}} = e^{-(\lambda_1 - \lambda_2)t} = e^{-(10\lambda - \lambda)t} = e^{-9\lambda t}$

Given, $\frac{N_1}{N_2} = e^{-1}$**. Hence** $-9\lambda t = -1$ **or** $t = \frac{1}{9\lambda}$**. Thus the correct choice is (d).**

Q.19 Answer:

30 min

Solution:

Since $\frac{1}{16} = \frac{1}{2^4}$**, it follows that 2 hours** $= 4$ **half lives. or 1 half-life** $= \frac{2}{4} = \frac{1}{2}$ **hour** $= 30$ **min. Hence the correct choice is (b).**

Q.20 Answer:

decay of a neutron in a nucleus

Solution:

Beta particle is an electron which is emitted from a nucleus when a neutron decays into a proton and an electron within a nucleus. Hence the correct choice is (c).

Q.21 Answer:

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Solution:

Activity $R = \lambda N$ and $\lambda = \frac{0.693}{T}$, where T is the half life of the sample. Now

$$T = \frac{0.693}{\lambda} = \frac{0.693N}{\lambda N} = \frac{0.693N}{R}$$

$$\text{Thus} \quad T_1 = \frac{0.693N_1}{R_1}$$

$$\text{and} \quad T_2 = \frac{0.693N_2}{R_2}$$

$$\therefore \quad \frac{T_1}{T_2} = \frac{N_1}{N_2} \times \frac{R_2}{R_1} = 2 \times 2 = 4$$

Q.22 Answer:

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Solution:

We know that $r_n \propto n^2$. Hence $\frac{r_1}{r_2} = \left(\frac{n_1}{n_2}\right)^2$. Given $r_1 = 4r_2$. Therefore

$$4 = \left(\frac{n_1}{n_2}\right)^2 \Rightarrow n_1 = 2n_2$$

Q.23 Answer:

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Solution:

Given $T = 20$ min. We know that

$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T}$

For 33% decay, $\frac{N}{N_0} = \frac{100-33}{100} = \frac{67}{100}$

If t_1 is the time for 33% decay, then

$\frac{67}{100} = \left(\frac{1}{2}\right)^{t_1/T}$ (i)

For 67% decay, $\frac{N}{N_0} = \frac{100-67}{100} = \frac{33}{100}$

If t_2 is the time for 67% decay, then

For 67% decay, $\frac{N}{N_0} = \frac{100-67}{100} = \frac{33}{100}$

If t_2 is the time for 67% decay, then

$\frac{33}{100} = \left(\frac{1}{2}\right)^{t_2/T}$ (ii)

Dividing (ii) by (i), we have

$\frac{33}{67} = \left(\frac{1}{2}\right)^{(t_2-t_1)/T}$ (iii)

Now $\frac{33}{67} \simeq \frac{1}{2}$. Hence Eq. (iii) reduces to

$\left(\frac{1}{2}\right)^1 = \left(\frac{1}{2}\right)^{(t_2-t_1)/T}$

which gives $1 = \frac{t_2-t_1}{T}$ or $t_2 - t_1 = T = 20$ min.

Q.24 Answer:

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Solution:

$r_n \propto n^2$. **Therefore** $n \propto \sqrt{r_n}$. **If** r_{i} **and** r_f **are the radii of the initial and final states, then**

$$n = \sqrt{\frac{r_f}{r_i}} = \sqrt{\frac{4.77}{0.53}} = \sqrt{9} = 3$$

Q.25 Answer:

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Solution:

$\frac{N}{N_0} = \mathrm{e}^{-\lambda t}$. **Thus** $\frac{1}{20} = \mathrm{e}^{-\lambda t}$ **or** $20 = \mathrm{e}^{\lambda t}$**or**

$\lambda t = \log_{\mathrm{e}}(20) = 3.0$

where $\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{6.93} = 0.1$ **per day**

Therefore $t = \frac{3.0}{\lambda} = \frac{3.0}{0.1} = 30$ **days** ≈ 1 **month.**

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